EQN

ALGEBRA FACTORING STUDIO & POLYNOMIAL TILES

Kinesthetic Quadratics, Factoring & Polynomial Binomial Arrays

SERIES 1 • CARD 06 / 12 • DOMAIN: POLYNOMIAL GEOMETRY & BINOMIAL FACTORING

CARD 06 / 12 • ALGB-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

GEOMETRIC AREA MODEL. MULTIPLY BY DIMENSION.

Quadratic equations model geometric area. Polynomial tiles model multiplication geometrically:

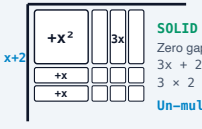
Area = Length × Width = (x + a)(x + b)

Arranging tiles into a rectangle factors polynomials!

CALIBRATED POLYNOMIAL DATUM

- Unit Datum: +1 Constant Tile = 1U = 7.00 mm pitch
- Linear Rod: +x Tile = 5U × 1U = 35.00 × 7.00 mm
- Quadratic: +x² Tile = 5U × 5U = 35.00 × 35.00 mm

AREA MODEL: (x + 3)(x + 2) = x² + 5x + 6



SOLID RECTANGLE

Zero gaps or overlaps

3x + 2x = 5x

3 × 2 = 6 units

Un-multiplying area!

2. KIT PARTS INVENTORY (43 PIECES)

COMPONENT	QTY	SPEC	COLOR
Quadratic (+x²)	2x	38.6×38.6 mm	Matte Blue
Quadratic (−x²)	2x	38.6×38.6 mm	Muted Red
Linear Rod (+x)	6x	38.6×19.0 mm	Matte Blue
Linear Rod (−x)	6x	38.6×19.0 mm	Muted Red
Constant (+1)	12x	19.0×19.0 mm	Matte Blue
Constant (−1)	12x	19.0×19.0 mm	Muted Red
Cartesian Rails	2x	Stage L-Rails	PLA White
3x1 Storage Bin	1x	49.0 mm	PLA White

TOTAL: 40 Polynomial Tiles + 2 L-Rails + 7U Box = 43 Parts

SLANT 3D PRODUCTION PALETTE

- PLA Matte Blue: Positive polynomial values (+x², +x, +1).
- PLA Matte Muted Red: Negative polynomial values (-x², -x, -1).
- PLA White: Storage enclosure body and Cartesian L-rails.
- Zero-Pair Elimination: Blue cancels red on sight.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Lay quadratic tiles flat on the box floor, slot linear rods in side bays, and stack unit cubes in center bays. L-rails snap into lid retention for rattle-free carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every degree-n polynomial has exactly n complex roots counted with multiplicity.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: FACTORING TRINOMIALS & BINOMIAL ARRAYS

Step-by-Step Hands-On Tutorial: Factoring $x^2 + 5x + 6 = (x+2)(x+3)$

SERIES 1 • CARD 06 OF 12 • DOMAIN: POLYNOMIAL GEOMETRY & BINOMIAL FACTORING

CARD 06 / 12 • ALGB-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/algebra-boxes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS HSA.SSE.A.2 (Structure)
- CCSS HSA.APR.A.1 (Arithmetic)
- TEKS 111.39.c.10.E (Factoring Tr...)

Grades 7–11 Pre-Algebra & Algebra II

GUIDING INQUIRY

Can you arrange the linear rods so the rectangle closes completely without any gaps or overlapping edges?

1 STEP 1: SEAT ORIGIN TILE (+x²)

Slide the two L-rails into the lid corner to form the origin.
Place the emerald +x² tile flush into the corner.
Its 35.00 × 35.00 mm area establishes the quadratic core.

2 STEP 2: DISTRIBUTE 5x LINEAR RODS

We need to distribute 5 positive x rods around the +x² tile.
Place 3 rods along the top rail (horizontal dimension).
Place 2 rods along the left rail (vertical dimension).

3–4 STEPS 3 & 4: COMPLETE RECTANGLE & SOLVE

3. COMPLETE THE RECTANGLE: Fill missing corner with 6 constant (+1) tiles!
4. READ BINOMIAL DIMENSIONS: Horizontal width is (x + 3); vertical length is (x + 2).
Total Area: $(x + 2)(x + 3) = x^2 + 5x + 6$!
Calculator Verification: Press Polynomial Solver Icon ([⁻¹][EQN] & [⁻¹][POLY]) and run [⁻¹]
[POLY] 1 [A] 5 [B] 6 [C] 0 [D] [XEQ] [RCL] 1 [RCL] 3 → -2.

5 CORE PRINCIPLE: FACTORING IS UN-MULTIPLYING AREA

Factoring a quadratic is literally finding the length and width of a rectangle! If the tiles cannot form a solid rectangle without gaps, the trinomial is prime. Zero-pairs (+x and -x) let you model differences of squares x^2 - 4 with effortless physical cancellation!

★ EXTENSION INVESTIGATION: FACTOR: x² + 6x + 8

Arrange one +x² tile, six +x rods, and eight +1 units into a solid rectangle. What are its rail dimensions? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

FDISP

FRACTION STUDIO PROPORTIONAL TOWERS

4-Way Multi-Notational Fraction Blocks (10 Denominators: 1/1 to 1/12)

SERIES 1 • CARD 04 / 12 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE

CARD 04 / 12 • FRAC-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

PROPORTIONAL HEIGHT & 4-WAY MULTI-NOTATION

Fractions are proportional lengths on a 7mm grid.

Each block face shows fraction, decimal, and percent:

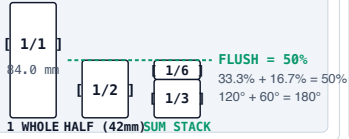
[1/3] (Symbol) = 0.333 (Decimal) = 33.3% (Percent)

Equal heights prove mathematical equivalence!

UNIVERSAL 84.00 MM WHOLE DATUM

- 1 Whole Datum: [1/1] = 12U = 84.00 mm reference rod
- Unit Base: [1/12] = 1U = 7.00 mm (Matches [2])
- Sub-Units: [1/2]=42mm, [1/3]=28mm, [1/4]=21mm

PROPORTIONAL EQUIVALENCE: 1/3 + 1/6 = 1/2



2. KIT PARTS INVENTORY (65 PIECES)

COMPONENT	QTY	SPEC	COLOR
Whole [1/1]	2x	84.00 mm	PLA Tan
Halves [1/2]	4x	42.00 mm	PLA Tan
Thirds [1/3]	3x	28.00 mm	Muted Red
Fourths [1/4]	4x	21.00 mm	Matte Blue
Fifths [1/5]	5x	16.80 mm	PLA White
Sixths [1/6]	6x	14.00 mm	Muted Red
Eighths [1/8]	8x	10.50 mm	Matte Blue
Ninths [1/9]	9x	9.33 mm	Muted Red
Tenths [1/10]	10x	8.40 mm	PLA White
Twelfths [1/12]	12x	7.00 mm	Matte Blue
13U Box & Lid	2x	98.4 mm	PLA Gray/White

TOTAL: 63 Blocks (12 Columns × 84mm) + 13U Box + Stage Lid = 65 Parts

SLANT 3D PRODUCTION PALETTE

- PLA White: Base-10 fractions (5x 1/5, 10x 1/10) & Stage Lid.
- PLA Matte Blue: Binary/duodecimal family (4x 1/4, 8x 1/8, 12x 1/12).
- PLA Matte Muted Red: Ternary fraction family (3x 1/3, 6x 1/6, 9x 1/9).
- PLA Tan: Unit wholes & halves (2x 1, 4x 1/2) • PLA Gray: 13U Box Body.

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Stand all 63 blocks across the 12 floor sockets (2 cols of 1, 2 cols of 1/2, and 1 col each of 1/3..1/12). Every column levels at 84.0 mm; missing pieces leave an instant

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Proportional height is strictly preserved under common denominator fraction addition.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: COMMON DENOMINATOR ADDITION & EQUIVALENCE

Step-by-Step Hands-On Tutorial: 1/3 + 1/4 = 7/12 & 4-Way Multi-Notation

SERIES 1 • CARD 04 OF 12 • DOMAIN: PROPORTIONAL TOWERS & MULTI-NOTATION EQUIVALENCE

CARD 04 / 12 • FRAC-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/fraction-studio/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 5.NF.A.1 (Unlike Denom...)
- CCSS 4.NF.A.2 (Fraction Com...)
- TEKS 111.7.b.3.H (Rational Ope...)

Grades 3–8 Fractions & Proportions

GUIDING INQUIRY

Which smaller block family levels off flush with both fraction towers in the comparison slot?

1 STEP 1: SEAT 1/1 WHOLE IN REFERENCE WELL

Place the 84.00 mm Whole [1/1] block in the left reference slot.
Its 4 faces show: 1/1, 1.00, 100%, and 360°.
This physical block establishes the 100% benchmark height.

2 STEP 2: STACK 1/3 AND 1/4 BLOCKS

Place one [1/3] block (28.00 mm) in the right comparison well.
Stack one [1/4] block (21.00 mm) directly on top of the [1/3].
Total combined height: 28.00 + 21.00 = 49.00 mm.

3–4 STEPS 3 & 4: TEST TWELFTHS & SOLVE

3. TEST COMMON TWELFTHS: Stack 7 twelfths: 7 × 7.00 mm = exactly 49.00 mm! Flush summit!
4. READ 4-WAY FACES: 0.333 + 0.250 = 0.583; 120° + 90° = 210°.
Calculator Verification: Press Fraction Display Icon ([FDISP] & [g][SIMP]) and run 1 [ENTER] 3 [÷] 1 [ENTER] 4 [÷] [+] [FDISP] → 7/12.

5 CORE PRINCIPLE: COMMON DENOMINATORS ARE COMMON SIZES

You cannot directly add thirds and fourths because their blocks are different physical heights!
Converting to twelfths (4/12 + 3/12 = 7/12) literally swaps disparate pieces for identical 7.00 mm unit blocks so their heights can be counted directly!

★ EXTENSION INVESTIGATION: ADD: 1/2 + 1/3

Seat [1/2] (42mm) and [1/3] (28mm) together. Test sixths: how many [1/6] blocks reach the exact 70.0mm summit? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

CLASSROOM PLANETARY CLOCK & INTERVAL SYSTEM

Kinesthetic Time, Epicyclic 12:1 Gears & Modular Arithmetic

SERIES 1 • CARD 03 / 12 • DOMAIN: KINEMATIC GEAR RATIOS & MODULAR ARITHMETIC

CARD 03 / 12 • TIME-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

EPICYCLIC 12:1 GEARS & MODULO ARITHMETIC

Time is continuous rotational motion & modular math.

StackCalc Clock features an epicyclic gearset:

12:1 Exact Planetary Reduction Ratio

When minute hand turns 360°, hour hand advances 30°!

CALIBRATED 60-MINUTE / 12-HOUR DATUM

- Minute Unit Datum: 1 Minute = 6° dial sector
- Hour Mark: 1 Hour Mark = 30° arc (12 total sectors)
- Quadrant: 1/4 Hour Wedge = 90° right-angle sector

PLANETARY GEARS: 12:1 RATIO AT 3:30 (105°)



12:1 EPICYCLIC GEARS

360° min → 30° hour

At 3:30: Hour hand = 105°

15-Min Quadrant Wedge:

90° arc = 1/4 of an hour

Modulo 12 & Modulo 60

2. KIT PARTS INVENTORY (18 PIECES)

COMPONENT	QTY	SPEC	COLOR
Clock Frame	1x	105.0 mm Dial	Slate Gray
12:1 Gears	1x	Planetary	Gold / Blue
15-Min Wedges	4x	90° Quadrant	Sky Blue
30-Min Wedges	2x	180° Sector	Royal Purple
5-Min Wedges	8x	30° Sector	Emerald Green
Stand & Lid	1x	Stand & Lid	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 16 Clock & Wedge Components + Storage Bin + Stand Lid = 18 Parts

FRACTION-ALIGNED TIME PALETTE

- Blue Wedges: 15-minute quadrants (90° right angles).
- Purple Wedges: 5-minute segments (30° hour steps).
- Emerald Face: Calibrated clock dial with 6° ticks.
- High-Visibility Hands: White minute & orange hour hands.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Seat the geared clock frame into its bottom cradle. Drop the 14 colored wedges into their labeled radial lid bays; any missing wedge leaves an instantly empty bay. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Clock arithmetic is modulo 12 and modulo 60 governed by strict continuous gear kinematics.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: ELAPSED TIME & PLANETARY GEAR RATIOS

Step-by-Step Hands-On Tutorial: 3:30 + 0:45 = 4:15 & Modular Time

SERIES 1 • CARD 03 OF 12 • DOMAIN: KINEMATIC GEAR RATIOS & MODULAR ARITHMETIC

CARD 03 / 12 • TIME–01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/classroom–clock/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 3.MD.A.1 (Elapsed Time)
- CCSS 4.MD.A.2 (Time Intervals)
- TEKS 111.5.b.7.C (Elapsed Time...)

Grades 2–6 Elapsed Time & Modular Math

GUIDING INQUIRY

How far does the hour hand travel when the minute hand turns one full circle around the dial?

1 STEP 1: SET START TIME TO 3:30

Rotate the minute hand to 6 (bottom of dial, 30 minutes).
Observe the internal gears: the hour hand advances smoothly halfway between 3 and 4 (exactly 105° on the dial).

2 STEP 2: INSERT 15–MINUTE WEDGES

We need to add 45 minutes of elapsed time.
Insert three 15-minute wedges (3 × 15 min = 45 min).
Wedges cover 3 × 90° = 270° angular arc around dial.
Notice: The wedges bridge past the 12

3–4 STEPS 3 & 4: ADVANCE GEARS & READ RESULT

3. ADVANCE MINUTE HAND: Turn hand clockwise across the 3 wedges past 4:00 to 4:15!
4. READ FINAL TIME: Hour hand at 4:15; minute hand at 3 (15 min).
Calculator Verification: Press Clock HMS Icon ([🕒] /[g][▸HMS] /[f][▸HR]) and run 3.30 [f] [▸HR] 0.45 [f] [▸HR] [+] [g] [▸HMS] → 4.15.

5 CORE PRINCIPLE: TIME IS CONTINUOUS KINEMATIC RATIO

Digital clocks make students think hour hands jump instantly when minutes reach 60. The planetary gears physically prove that time is a continuous ratio! Because 45 minutes is 3/4 of an hour, the hour hand moves through exactly 3/4 of its 30° sector (22.5°), landing precisely at

★ EXTENSION INVESTIGATION: ELAPSED TIME: 8:45 + 1:30

Set clock to 8:45. Add two 45-minute wedges (90 min). Where does the planetary gear place the hour hand? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

TEN-FRAMES & PLACE VALUE ARITHMETIC LAB

5-Anchor Subitizing, Regrouping & Base-10 Manipulatives

SERIES 1 • CARD 01 / 12 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING

CARD 01 / 12 • BASE-10

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

5-ANCHOR SUBITIZING & BRIDGING TO 10

Ten-frames visually organize numbers into 5-anchors:

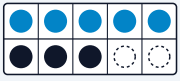
5-Anchor Formula: 5 + n • Ten's Comp: 10 - n

Seeing 8 as 5 on top and 3 below anchors addition,
turning counting into instant visual recognition!

CALIBRATED 14.00 MM SUBITIZING GRID

- Counter Diameter: 14.00 mm (2U) with two-tone face
- Stacking Datum: 1.85mm base plug & stacking socket
- Ten-Frame Grid: 2 × 5 array with 14.00 mm wells

5-ANCHOR SUBITIZING: 5 + 3 = 8 (BRIDGE 10)



INSTANT SUBITIZING

Top Row: 5 Blue (anchor)
Bottom: 3 Red (8 total)

2 Empty Slots:

10 - 8 = 2 Ten's Comp
Bridge: 8 + 5 = 13

2. KIT PARTS INVENTORY (56 PIECES)

COMPONENT	QTY	SPEC	COLOR
10-Frame Trays	2x	97.4×38.6mm	PLA Gray
Counter Blocks A	20x	19.0 mm Tile	PLA White
Landmark [10] A	6x	19.0 mm Tile	PLA White
Counter Blocks B	20x	19.0 mm Tile	PLA Tan
Landmark [10] B	6x	19.0 mm Tile	PLA Tan
Twin Stage Lid	1x	Twin 10-Frame	PLA White
3x1 Storage Bin	1x	49.0 mm	PLA White

TOTAL: 40 Counters + 12 Landmark [10] Blocks + 2 Trays + Box + Lid = 56 Parts

SLANT 3D PRODUCTION PALETTE

- PLA White: Set A (20x Counters + 6x [10] Blocks), Box Body & Stage Lid.
- PLA Tan: Set B (20x Counters + 6x [10] Blocks) for second operand.
- PLA Gray: Dual 5x2 Ten-Frame activity trays for high contrast.
- All 52 blocks pack in 4 columns of 13 blocks + 2 side trays.

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Populate both 10-frames with 10 counters each and lay carry cubes into center trough. Empty wells instantly spot missing pieces. Twin stage lid snaps flush.

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Base-10 positional notation decomposes any integer into powers of 10 with digits 0 through 9.

LESSON GUIDE: MAKE-A-TEN BRIDGE & 5-ANCHOR SUBITIZING

Step-by-Step Hands-On Tutorial: 8 + 5 = 13 via Bridging to 10

SERIES 1 • CARD 01 OF 12 • DOMAIN: PLACE VALUE & 5-ANCHOR SUBITIZING

CARD 01 / 12 • BASE-10

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/ten-frames/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Make Ten)
- CCSS 2.NBT.B.5 (Place Value)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K–3 Subitizing & Place Value

GUIDING INQUIRY

How many empty spots are left in Frame A before you start moving counters from Frame B?

1 STEP 1: POPULATE 8 ON TEN-FRAME A

Place 8 blue counters in Ten-Frame A.
Fill the top row (5 blue) and bottom row (3 blue).
Instant Subitizing: 5 + 3 = 8 (zero counting needed!).
Notice: exactly 2 slots remain empty to

2 STEP 2: POPULATE 5 ON TEN-FRAME B

Place 5 red counters in Ten-Frame B (fills top row of 5).
We want to calculate 8 + 5 without counting by ones.
Frame A needs 2 counters to bridge to 10.

3–4 STEPS 3 & 4: BRIDGE TO 10 & REGROUP

3. BRIDGE TO 10: Move 2 red counters from B to A. Frame A is now 1 Ten (10)!

4. READ REMAINING ONES: Frame B has 3 red counters left. Total: 10 + 3 = 13!

Keystrokes: Ten-Frame Icon ([▤] & [PV]) 8 [ENTER] 2 [+] 5 [+] [▤] → 15.

5 CORE PRINCIPLE: ADDITION IS BRIDGING, NOT COUNTING

Counting on fingers by ones is slow and error-prone. The 10-frame proves that addition is decomposing numbers into friendly anchors: 8 + 5 = 8 + (2 + 3) = (8 + 2) + 3 = 10 + 3 = 13! Once students see the 2 empty spaces in the 8-frame, they instantly know to borrow 2 from the 5!

★ EXTENSION INVESTIGATION: BRIDGE TO 10: 7 + 6

Put 7 in Frame A (3 empty). Put 6 in Frame B. Move 3 to make a ten. How many are left in B? 10 + 3 = 13! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

PLACE VALUE STUDIO & REGROUPING TRAYS

Dual Whole Number & Decimal Activity Trays with Base-Ten Regrouping Tokens

SERIES 1 • CARD 12 / 12 • DOMAIN: POSITIONAL BASE-10 & REGROUPING

CARD 12 / 12 • PLACE-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

POSITIONAL BASE-10 NOTATION & REGROUPING

A digit's value depends on its place column: $d_k \times 10^k$.

Addition: Trade 10 units for 1 Carry block (+1) one column left.

Subtraction: Borrow 1 from left column for a (10) Trade block.

Dual 5x2 trays connect whole numbers directly to decimals!

CALIBRATED 19.00 MM POSITIONAL GRID

- Digit Tile: 19.00 × 19.00 × 7.00 mm (1U) with 1.85mm detent
- Whole Tray: 10,000, 1000, 100, 10, 1 (Inner 5×2 Tray #1)
- Decimal Tray: 0.1, 0.01, 0.001, 10^{-4}, 10^{-5} (Inner 5×2 Tray #2)

PLACE VALUE REGROUPING: 347 + 285 = 632

THOU	HUND	TENS	ONES
		+1	
	3	4	7
	2	8	5

REGROUPING TOKENS

Trade 10 Ones for 1 Ten

7 + 5 = 12 → 2 ones, +1 carry

Dual 6x2 Trays:

Whole: Thousands to Ones

Decimal: Ones to Thousandths

2. KIT PARTS INVENTORY (56 PIECES)

COMPONENT	QTY	SPEC	COLOR
White Digits (0–9)	22x	4×0, 2×1–9	PLA White
Blue Digits (0–9)	22x	4×0, 2×1–9	Blue-Gray
Carry & Borrow Tokens	8x	4×, 4×	Glow Yellow
Whole Numbers Tray	1x	Inner 5×2	Blue-Gray
Decimal Numbers Tray	1x	Inner 5×2	Blue-Gray
3x1 Storage Bin	1x	49.0 mm	PLA White
[123] Docking Lid	1x	Docking Lid	PLA White

TOTAL: 44 Digit Tiles + 8 (+1/10) Tokens + 2 Inner 5x2 Trays + Box + Lid = 56 Parts

SYMMETRIC 2-COLOR OPERAND & REGROUPING PALETTE

- PLA White (Set A, 22x): 4x '0' + 2x each of '1'–'9' + Box + Lid.
- Slate Blue-Gray (Set B, 22x): 4x '0' + 2x each of '1'–'9'.
- Slate Blue-Gray (2x Inner Trays): 5x2 Whole & Decimal Trays.
- Glow Yellow / Red (8x): 4x (+1) Carry + 4x (10) Borrow tokens.

3. PACK-UP & SUPPORT

✓ **10-SECOND PACK-UP**

Stack 4 blocks in Cols 0–4 (0–9), stack 6 blocks in Col 5 (+1, 10, & extra 0s = 52 blocks total), nest the 2 inner 5x2 trays over Cols 0–4 (leaving Col 5 for easy finger grip)

✓ **NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ **HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every real number has a unique base-10 positional representation.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: MULTI-DIGIT ADDITION & DECIMAL REGROUPING

Step-by-Step Hands-On Tutorial: 347 + 285 = 632 with Regrouping Tokens

SERIES 1 • CARD 12 OF 12 • DOMAIN: POSITIONAL BASE-10 & REGROUPING

CARD 12 / 12 • PLACE-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/place-value/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 2.NBT.A.1 (Place Value)
- CCSS 4.NBT.B.4 (Addition/Sub...)
- CCSS 5.NBT.A.3 (Decimals to ...)

Grades 1–5 Elementary Mathematics

GUIDING INQUIRY

What happens when the counters in any column reach ten?

1 STEP 1: POPULATE OPERANDS 347 AND 285

Place 347 in top row (White) and 285 in bottom row (Tan).
Align 3 hundreds, 4 tens, 7 ones over 2 hundreds, 8 tens, 5 ones.
Visual clarity: operand contrast eliminates confusion.

2 STEP 2: REGROUP ONES (7 + 5 = 12)

7 + 5 = 12. Trade 10 ones for a Red (+1) token.
Place the Red token on the Tens column digit block.
Keep 2 in the Ones column.

3–4 STEPS 3 & 4: REGROUP TENS & COMPLETE

3. REGROUP TENS: 4 + 8 + 1 = 13. Trade 10 tens for a Red (+1) in Hundreds.
4. READ COMPLETED SUM: Keep 3 in Tens; Hundreds: 3 + 2 + 1 = 6. Total = 632!

5 CORE PRINCIPLE: CARRYING IS PHYSICAL TRADING

In standard algorithms, carrying is a mysterious tiny 1 written above a column.
Here, carrying is a physical trade: 10 units become 1 ten block.
Students see that value is conserved across columns!

★ EXTENSION INVESTIGATION: DECIMAL CHALLENGE: 1.75 + 0.85

Set up Decimal tray with 1.75 and 0.85. Regroup hundredths and tenths. What is the sum?
2.60! (Scan QR for 3D lesson!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

PLACE VALUE STUDIO — PACKING BENCH & QC SPEC

Symmetric 22 White + 22 Blue-Gray Digits, 4x (+1) Carry, 4x (10) Borrow & Dual 5x2 Trays

WAREHOUSE FULFILLMENT & PACKING BENCH REFERENCE • REV a793ddd • 3x1x7U Box + 2 Inner 5x2 Trays + [123] Lid

SKU: PLACE-01-PACK

REV a793ddd

PACKING BENCH MANIFEST & REFILL REPLACEMENT PRICING

#	COMPONENT DESCRIPTION	QTY	COLOR / SPEC	CONTENT-ADDRESSED REF	REFILL
①	**White Digits A (4×0, 2×1–9)**	**22x**	PLA White	**PLAC-DIG-WHT**	**$1.88**
②	**Blue Digits B (4×0, 2×1–9)**	**22x**	Blue-Gray / Tan	**PLAC-DIG-BLU**	**$1.88**
③	**4× [+1] Carry & 4× [10] Borrow**	**8x**	Glow Yellow	**PLAC-TOKN-RG**	**$2.25**
④	**Inner 5x2 Whole Tray (L5)**	**1x**	Blue-Gray / Gray	**PLAC-TRAY-WHL**	**$6.75**
⑤	**Inner 5x2 Decimal Tray (L6)**	**1x**	Blue-Gray / Gray	**PLAC-TRAY-DEC**	**$6.75**
⑥	**3x1x7U Box (12 Sockets) & Lid**	**2x**	PLA White / Gray	**PLAC-BOX-7U**	**$19.50**

TOTAL: 56 VERIFIED WORKING COMPONENTS

WAREHOUSE NO LOST PIECE GUARANTEE

Individual replacement blocks are permanently stocked in Mountain View, California.
Anti-arbitrage pricing: Order single replacement components anytime at stackcalc.io/parts.
Never retire an educational classroom kit due to a single misplaced component.

PACKING QC & WATERLINE PROTOCOL

FINAL BENCH INSPECTION

- ✓ 22x White Digits: 4x [0] + 2x [1]–[9]
- ✓ 22x Blue Digits: 4x [0] + 2x [1]–[9]
- ✓ 4x [+1] Carry + 4x [10] Borrow = 8 tokens
- ✓ 2x Inner 5x2 Trays (Whole + Decimal)

PACKING WATERLINE NOTICE

56-PC PACK-UP: Seat 4 blocks in Cols 0–4
(0–9 = 40 pcs), 6 blocks in Col 5 (4x [0],
4x [+1], 4x [10] = 12 pcs), nest both 5x2
Trays over Cols 0–4, and snap Lid shut.

CALIFORNIA LAB QC AUDIT

INSPECTION: PASSED

Inspector: Station 04 • Mountain View, CA
Lot Release: 2026-Q4-ARCHIVAL
Standards: ASTM F963 / EN71 / RoHS Lead-Free

FACTORY SPECIFICATION
Harmonized Modular Mathematics System • Manufactured with Pride in California, USA.

INSTITUTIONAL DIRECT
sales@stackcalc.io

PLACE VALUE STUDIO — PACKING BENCH & QC SPEC

WAREHOUSE PACKING BENCH REFERENCE • 3D VIDEO QR • REFILL LOGISTICS

Classroom Pack-Up & Inventory Reference

SKU: PLACE-01-PACK

PACKING PROTOCOL

PACK-UP DEMO VIDEO

OPEN CAMERA & SCAN

stackcalc.io/products/place-value/

Watch 3D video demonstration

WAREHOUSE PACKING BENCH & BOX FLOOR WATERLINE AUDIT

1. STACK 40 DIGITS IN COLS 0–4 (LAYERS 1–4)

Place 4 blocks (2 White + 2 Blue-Gray) into each of the 10 digit sockets (Row 1: 0–4; Row 0: 5–9).

2. NEST BOTH 5x2 TRAYS OVER COLS 0–4 (LAYERS 5–6)

Seat the 5x2 Whole Numbers Tray (Layer 5) and 5x2 Decimal Tray (Layer 6) over Columns 0–4.

3. STACK 12 BLOCKS IN COL 5 (LAYERS 1–6) & SNAP LID

Stack the 4 extra '0' blocks + 4x [+1] Carry + 4x [10] Borrow blocks 6-high in Column 5 and snap [123] Lid shut.

FULFILLMENT PORTAL

Interactive Packaging Bench Station • Warehouse Fulfillment Rev a793ddd.

REPLACEMENT MANIFEST

stackcalc.io/parts

MATH DOMINO STUDIO & 12-FRAME REGROUPING LAB

Triangular Combinatorics, Subitizing & Part-Whole Decompositions

SERIES 1 • CARD 02 / 12 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS

CARD 02 / 12 • DOMI-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

TRIANGULAR COMBINATORICS & PART-WHOLE PAIRS

Double-six dominoes form a triangular math system.

It contains all unordered pairs [alb] from 0 to 6:

Total Tiles = T(7) = (7 × 8) / 2 = 28

Unique Tiles

Each tile is a 2-part additive partition of 0–12!

MODULAR 28.0 × 14.0 MM TILE DATUM

- Tile Size: 28.00 × 14.00 × 7.00 mm (4U×2U×1U)
- Tactile Pips: 3.20 mm ⌀ recessed high-contrast cups
- Center Divider: Recessed 1.2mm scoring groove

DOMINO SUBITIZING & SUM: [5|4] + [3|2] = 14



[5|4] = 9 [3|2] = 5

PART-WHOLE SUM

[5|4] pips = 9

[3|2] pips = 5

Total: 9 + 5 = 14

T(7) = 28 Unique Tiles

Double-Six Combinatorics

2. KIT PARTS INVENTORY (33 PIECES)

COMPONENT	QTY	SPEC	COLOR
Double-Six Tiles	28x	28×14×7 mm	Stealth Black
Wildcard Blanks	2x	28×14×7 mm	Stealth Black
12-Frame Tray	1x	Dual 6-Bay	Matte Charcoal
Doubles Lid	1x	7-Slot Rail	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 30 Domino Tiles + 12-Frame Tray + Doubles Lid + Storage Bin = 33 Parts

HIGH-CONTRAST PIPS & TEXTURE

- Stealth Black Matte: Low glare, fingerprint resistant polymer.
- Brilliant White Pips: Standard dice quincunx configurations.
- Doubles Spine: Labeled slots [0|0] to [6|6] for organizing.
- High Durability: Scratch-resistant tactile surfaces.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Arrange the 28 standard tiles and 2 wildcards into 4 columns of 7 and 1 row of 2 on the floor. Missing tiles break the brick layout! Snap lid securely.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

The number of unordered pairs chosen with replacement from n elements is T(n) = n(n+1)/2.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: TRIANGULAR COMBINATORICS & PART–WHOLE SUMS

Step-by-Step Hands-On Tutorial: T(7) = 28 Tiles & [5|4] + [3|2] = 14

SERIES 1 • CARD 02 OF 12 • DOMAIN: DISCRETE COMBINATORICS & PARTITIONS

CARD 02 / 12 • DOMI–01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/math–dominoes/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS K.OA.A.3 (Decompose to 10)
- CCSS 1.OA.B.4 (Unknown Addend)
- TEKS 111.3.b.3.A (Model Additi...)

Grades K–5 Addition & Combinatorics

GUIDING INQUIRY

If you flip a domino end-for-end, does the total pip sum change? Why or why not?

1 STEP 1: LINE UP DOUBLES SPINE [0|0] TO [6|6]

Place the 7 double tiles in the lid's organizer spine:
[0|0]=0, [1|1]=2, [2|2]=4, [3|3]=6, [4|4]=8, [5|5]=10, [6|6]=12.
These 7 doubles form the diagonal spine of the 28-tile triangle!

2 STEP 2: PAIR COMPLEMENTARY SUM–TO–6 TILES

Find tiles whose pip sum equals 6: [0|6], [1|5], [2|4], [3|3].
Place them in the 12-frame tray side by side.
Observe the inverse symmetry: as left side increases by 1,

3–4 STEPS 3 & 4: COMBINE TILES & REGROUP

3. COMBINE TWO TILES: Take tile [5|4] (sum 9) and [3|2] (sum 5). Seat in 12-frame tray.
4. CALCULATE SUM: 9 + 5 = 14! (1 full 10-frame + 4 ones).
Calculator Verification: Press Domino Pip Pattern Icon ([g][COUNT] & [g][a:b]) and run 5 [ENTER] 4 [+] 3 [ENTER] 2 [+] [+] → 14.

5 CORE PRINCIPLE: THE 28–TILE TRIANGULAR NUMBER SECRET

Why are there exactly 28 tiles? Count them by their highest pip: 7 tiles have a 0 ([0|0]..[0|6]),
6 new tiles have a 1 ([1|1]..[1|6]), 5 have a 2, down to 1 tile with a 6 ([6|6]).
Sum: 7 + 6 + 5 + 4 + 3 + 2 + 1 = 28! Dominoes are triangular numbers made tactile!

★ EXTENSION INVESTIGATION: FIND ALL SUM–TO–7 TILES

Can you find all 3 tiles that sum to 7: [1|6], [2|5], and [3|4]? Why is there no [0|7] tile in a double-six set? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

PROBABILITY DICE & MODULAR SPINNER PACK

Random Variables, Sample Spaces & Statistical Distributions

SERIES 1 • CARD 07 / 12 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS

CARD 07 / 12 • PROB-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

SAMPLE SPACES, 2d6 SUMS & NORMAL CURVES

Randomness follows strict combinatorial laws.

Rolling two fair 6-sided dice gives 36 outcomes:

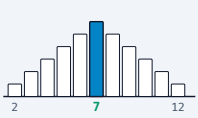
Sample Space = 6 × 6 = 36 Pair Outcomes

2d6 sum forms a triangle distribution peak at 7!

CALIBRATED 16.00 MM FAIR DICE DATUM

- Fair d6: 16.00 mm precision cube, balanced pips
- Polyhedral d20: 20-sided icosahedron (1/20 odds)
- Low-Friction Spinner: Precision brass bushing pivot

2d6 SUM PROBABILITY: TRIANGULAR PEAK AT 7 (6/



PEAK AT 7 (16.7%)

6 / 36 Combinations

Symmetric Triangle

Sample Space: 6×6=36

Birth of the Bell Curve

2. KIT PARTS INVENTORY (14 PIECES)

COMPONENT	QTY	SPEC	COLOR
Fair d6 Dice	4x	16.0 mm Fair	High-Contrast
Polyhedral d20	1x	20-Sided d20	Emerald Green
Precision Hub	1x	Brass Hub	High-Spin Brass
Spinner Dials	4x	3 Dials Set	Polymer Plates
Sample Grid Lid	1x	6×6 Matrix	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 10 Working Parts + 2 Extra Dials + Storage Bin + Lid = 14 Parts

PROBABILITY DISTRIBUTION PALETTE

- 4 Color-Separated d6 Dice: Enables independent event tracking.
- Emerald d20 Die: High-visibility prime distribution testing.
- High-Spin Brass Bushing: Ultra-low friction for random spins.
- Clear Calibration: High-contrast legible numerals.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Seat the 4 d6 dice and 1 d20 into bottom bays. Stack spinner dials flat and click brass hub onto post. Empty bays show missing pieces instantly. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

The sum of independent random variables tends toward a normal distribution as sample size grows.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: 2d6 SAMPLE SPACES & INDEPENDENT EVENTS

Step-by-Step Hands-On Tutorial: Why 7 is the Most Likely Roll (P = 6/36)

SERIES 1 • CARD 07 OF 12 • DOMAIN: SAMPLE SPACES & PROBABILITY DISTRIBUTIONS

CARD 07 / 12 • PROB-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/dice-and-spinner/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 7.SP.C.8 (Compound Events)
- CCSS HSS.CP.A.2 (Independence)
- TEKS 111.27.b.6.B (Representati...)

Grades 5–10 Probability & Statistics

GUIDING INQUIRY

Why are there six different combinations to roll a 7, but only one combination to roll a 2?

1 STEP 1: INSPECT THE 6 × 6 SAMPLE SPACE

Examine the 6 × 6 grid on the lid stage.
Die 1 rows (1-6) × Die 2 columns (1-6) =
36 total outcomes.
Every individual pair (like [3,4] or [6,1]) has probability 1/36.

2 STEP 2: COUNT PATHS TO EACH SUM

Trace diagonals of constant sum across the grid:
• Sum 2: Only 1 way [1,1] → P = 1/36 (2.8%)
• Sum 7: 6 ways [1,6],[2,5],[3,4],[4,3],[5,2],[6,1] → P =

3–4 STEPS 3 & 4: SAMPLE TRIALS & CALCULATE

3. SAMPLE EXPERIMENTAL RUNS: Roll 2d6 three times: 7, 4, 10.

4. COMPUTE SAMPLE MEAN: Calculate mean on calculator:

Press Dice & Statistics Icon (, [Σ+] & [g][x̄]) and run [g] [CLΣ] 7 [Σ+] 4 [Σ+] 10 [Σ+] [g] [x̄] → 7.

Independent Event: d6=6 and Spinner=10 → 1 [ENTER] 6 [÷] 1 [ENTER] 10 [÷] [×] [FDISP] → 1/60

5 CORE PRINCIPLE: THE BIRTH OF THE BELL CURVE

Single dice have flat uniform distributions (each face is 1/6). But when two dice add together,
extreme sums require extreme coordination, while central sums have many different additive combinations!
This combinatorial clustering is the foundation of the normal distribution Bell Curve!

★ EXTENSION INVESTIGATION: EVEN SUM PROBABILITY

Count all 2d6 combinations that give an EVEN sum (2, 4, 6, 8, 10, 12). Exactly 18 out of 36 outcomes! P = 50.0%! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Special Right Triangles, Radians, Coordinates & Wave Projections

SERIES 1 • CARD 08 / 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

CIRCULAR COORDINATES & DYNAMIC PROJECTIONS

Trigonometry is circular motion, not triangles.

On unit circle (r = 1), coordinates are trig values:

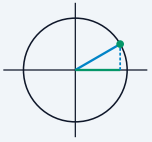
(X, Y) = (cos θ, sin θ)

Rotating disc projects angle θ onto cos/sin!

CALIBRATED 84.00 MM RADIUS DATUM

- Outer Ring: 84.00 mm ⌀ (Matches Fraction 1 Whole)
- Dual Scales: Degrees (0°–360°) and Radians (0–2π)
- Special Angles: Exact cutouts for 30°, 45°, and 60°

UNIT CIRCLE: 30° –> (cos = √3/2 ≈ 0.866, sin



ANGLE θ = 30° (π/6)

X = cos(30°) = √3/2 ≈ 0.8

Y = sin(30°) = 1/2 = 0.50

Pythagorean Identity:

cos²(θ) + sin²(θ) = 1

0.75 + 0.25 = 1.000

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC	COLOR
Unit Circle Disc	1x	84.0 mm ⌀	PLA White
Radian Overlay	1x	Radial Cursor	Clear PETG
Projection Arm	1x	Windows	Clear PC
Center Pivot	1x	Brass Bushing	Solid Brass
Stand & Lid	1x	Desk Stand	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal

TOTAL: 4 Volvelle Components + Storage Bin + Stand Lid = 6 Parts

QUADRANT SIGN CONVENTION

- Quadrant I (+, +): All functions positive (0° to 90°).
- Quadrant II (-, +): Sine positive, Cosine negative (90°–180°).
- Quadrant III (-, -): Tangent positive, Sine/Cos negative.
- Quadrant IV (+, -): Cosine positive, Sine negative.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Align transparent cursor to 0° / 360°, slip the volvelle into its protective sleeve, and lay flat into the box recess. Snap the stand lid shut for silent, rattle-free carry.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

For any real angle θ on the unit circle: cos²(θ) + sin²(θ) = 1 identically.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

Step-by-Step Hands-On Tutorial: Angle 30° -> (cos 30° = √3/2, sin 30° = 1/2)

SERIES 1 • CARD 08 OF 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/unit-circle-volvelle/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS HSF.TF.A.2 (Unit Circle)
- CCSS HSF.TF.A.3 (Special Tria...)
- TEKS 111.42.c.2.P (Trig Functio...)

Grades 9–12 Trigonometry & Physics

GUIDING INQUIRY

When the cursor rotates past 90° into Quadrant II, why does the horizontal X coordinate become negative while Y stays positive?

1 STEP 1: ROTATE CURSOR TO 30° (π/6 RAD)

Turn the transparent cursor arm to the 30° mark.
Look at the inner radian scale: 30° aligns exactly with π/6 rad.
Notice the right triangle projected onto the unit circle base!

2 STEP 2: READ HORIZONTAL COSINE PROJECTION

Look down at the horizontal X-axis window:
$X = \cos(30°) = \sqrt{3}/2 \approx 0.8660$.
This is the horizontal base length of the 30°-60°-90° triangle.

3–4 STEPS 3 & 4: READ SINE & VERIFY ON STACKCALC

3. READ VERTICAL SINE: Look at vertical Y-axis window: Y = sin(30°) = 1/2 = 0.5000.
4. VERIFY PYTHAGOREAN IDENTITY: $\cos^2(30°) + \sin^2(30°) = 0.75 + 0.25 = 1.0000$!
Calculator Verification: Press Unit Circle Radian Icon ([COS], [SIN] & [⁻][‣y,x]) and run 45 [COS] [⁻¹] [x²] 45 [SIN] [⁻¹] [x²] [+] → 1.

5 CORE PRINCIPLE: SINE AND COSINE ARE CIRCLE COORDINATES

Textbook right triangles only work for angles < 90°. The volvelle proves that sine and cosine
are the literal (X, Y) coordinates of a point moving around a circle! In Quadrant II, cosine naturally becomes negative because X is negative, while sine remains positive!

★ EXTENSION INVESTIGATION: SPECIAL ANGLE: 60° (π/3 RAD)

Rotate cursor to 60°. Notice X and Y coordinates swap: $X = \cos(60°) = 1/2 = 0.500$, $Y = \sin(60°) = \sqrt{3}/2 \approx 0.866$! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

EQN

STACKCALC SC-32 SCIENTIFIC RPN CALCULATOR

4-Register RPN Stack Engine, Dual Stand Modes & Hardware Guide

SERIES 1 • CARD 09 / 12 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES

CARD 09 / 12 • SC-32-HW

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

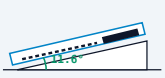
4-REGISTER RPN ENGINE & DUAL STANDS

Reverse Polish Notation (RPN) with 4 registers.
Numbers push onto stack; operators evaluate at once:
Stack Architecture: Regs T, Z, Y, X
Zero parentheses, zero ambiguity, zero errors.

CALIBRATED 32-KEYCAP TACTILE HARDWARE

- Display: 128×64 OLED with live 4-stack view
- Keycaps: 43 sculpted Shore 90A TPU keys over tactile switches
- Stack: Push on [ENTER], pop on op, LastX saved

OPERATING MODES: 11.6° DESK INCLINE & 78° PUZ



DESK STAND (11.6°)
Seated typing • Glare-free OLED



PUZZLE STAND (78°)
3-pc easel • Knocks flat to 6mm

2. KIT PARTS INVENTORY (6 PIECES)

COMPONENT	QTY	SPEC	COLOR
SC-32 Calculator	1x	128×64 OLED	Obsidian
TPU Bumper Sleeve	1x	Shore 95A TPU	Black TPU
3-Piece Stand	1x	78° Easel	Matte Charcoal
Desk Stand Caddy	1x	11.6° Incline	Matte Charcoal
USB-C Cable	1x	1.0m Braided	Stealth Black
Mailer Box	1x	6×4 In Case	Kraft Box

TOTAL: Calculator + TPU Sleeve + Stands + Cable + Box = 6 Parts

POWER CONTROL & ANNUNCIATORS

- Instant Power Off: Tap [f] then [ON/OFF] to shut down.
- Auto-Sleep Standby: Sleeps after 5 min idle (<15 μA).
- Non-Volatile Memory: Registers (X, Y, Z, T) persist.
- Annunciators: RAD/DEG, [f], [g], battery on OLED header.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Press [f] [OFF] to turn off, or let auto-sleep preserve your stack. Dock in the Desk Stand or slip into the TPU bumper sleeve and drop in your backpack with zero loose parts!

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Any algebraic expression can be evaluated unambiguously without parentheses using a postfix stack.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: 4-REGISTER RPN STACK & EXPRESSION TREES

Step-by-Step Hands-On Tutorial: Pythagorean Hypotenuse $\sqrt{(3^2 + 4^2)} = 5$

SERIES 1 • CARD 09 OF 12 • DOMAIN: 4-REGISTER RPN COMPUTING & EXPRESSION TREES

CARD 09 / 12 • SC-32-HW

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/sc32-calculator/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS MP5 (Use Tools St...)
- CCSS MP6 (Attend to Precision)
- TEKS 111.39.c.1.C (Manipulative...)

Grades 8–16 STEM, Physics & Engineering

GUIDING INQUIRY

Where does the first number wait safely on the stack while you are entering the second number?

1 STEP 1: ENTER 3 AND 4 ONTO STACK

Power on the SC-32. Key: 3 [ENTER] 4 [ENTER].
Stack state: Register Y holds 3.0000; Register X holds 4.0000.
In RPN, numbers always enter before operators!

2 STEP 2: SQUARE BOTH REGISTERS

Press [⁻¹] [x²] on 4: Register X becomes 16.0000.
Press swap [x⇄y]: Register X becomes 3.0000; Y holds 16.0000.
Press [⁻¹] [x²] on 3: Register X becomes 9.0000.

3–4 STEPS 3 & 4: COLLAPSE STACK & SQRT

3. ADD SQUARES: Press [+]. The stack drops: 16.0000 + 9.0000 = 25.0000 in X!
4. SQUARE ROOT: Press [√x]. Result: 5.0000!
Combined Keystrokes: 3 [⁻¹] [x²] 4 [⁻¹] [x²] [+] [√x] → 5.

5 CORE PRINCIPLE: THE FREEDOM FROM PARENTHESES

Evaluating $\sqrt{(3^2 + 4^2)}$ in traditional algebraic mode requires nested parentheses. In RPN, you calculate from the inside out, exactly as you think! Intermediate numbers sit patiently on the stack until an operator consumes them. Zero parentheses, zero syntax errors!

★ EXTENSION INVESTIGATION: KINETIC ENERGY: $E = \frac{1}{2} m v^2$

Calculate kinetic energy for mass m=12 and velocity v=5: 0.5 [ENTER] 12 [×] 5 [⁻¹] [x²] [×]
→ 150.0000! Zero parentheses! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

PRIME FACTOR STUDIO — IN-BOX USER GUIDE

Kinesthetic Number Theory, 4-3-3-2 Eratosthenes {2,3,5,7} Set & 12-Socket Lid Tray

SERIES 1 • CARD 05 / 12 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC

CARD 05 / 12 • PRIME-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

FEEL THE PRIMES. MULTIPLY WITH HEIGHT.

Standard base-10 counting blocks add lengths (1+1=2).

Prime Factor blocks are logarithmically proportional:

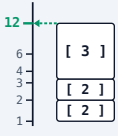
log(a × b) = log(a) + log(b)

Stacking blocks in a tower multiplies their numbers!

CALIBRATED LOGARITHMIC DATUM

- Unit Datum: Block [2] = 1U = 7.00 mm visible height
- Log Modulus: K_log = 23.2535 mm / decade
- Prime Formula: h_vis(p) = K_log × log10(p)

VISUAL PROOF: TOWER 12 = [2] + [2] + [3]



▲ SUMMIT = 12

2 × 2 × 3 = 12

Visible: 25.09 mm

Commutative:

[3]+[2]+[2] = same height!

2. KIT PARTS INVENTORY (45 PIECES)

COMPONENT	QTY	SPEC	COLOR
Prime Block [2]	**24x**	7.00 mm	Matte Blue
Prime Block [3]	**9x**	11.10 mm	Muted Red
Prime Block [5]	**6x**	16.25 mm	PLA White
Prime Block [7]	**4x**	19.65 mm	PLA Tan
7U Stepped Box	**1x**	56.4 mm	PLA Gray
12-Socket Lid	**1x**	125.5×42.0mm	PLA White

TOTAL: 43 Pure Prime Blocks (4-3-3-2 Layout) + 7U Box + Stage Lid = 45 Parts

SLANT 3D PRODUCTION PALETTE

- PLA Matte Blue: 24x Prime [2] (4 columns × 6 blocks = 42.0mm).
- PLA Matte Muted Red: 9x Prime [3] (3 columns × 3 blocks + 8.71mm riser).
- PLA White: 6x Prime [5] (3 columns × 2 blocks + 9.49mm riser) & Stage Lid.
- PLA Tan: 4x Prime [7] (2 cols × 2 blocks + 2.70mm riser) • PLA Gray: 7U Box.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Stand all 43 prime blocks in the 4-3-3-2 stepped floor sockets (4 cols of [2], 3 cols of [3], 3 cols of [5], 2 cols of [7]). All 12 columns level at the 42.0mm (2^6=64) waterline. Snap

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Every integer > 1 has a unique prime factorization decomposition.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

LESSON GUIDE: HOW TO FIND THE LEAST COMMON MULTIPLE (LCM)

Step-by-Step Hands-On Tutorial: Finding LCM(12, 18) = 36 & The Law of Conservation of Primes

SERIES 1 • CARD 05 OF 12 • DOMAIN: FUNDAMENTAL THEOREM OF ARITHMETIC

CARD 05 / 12 • PRIME-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/prime-studio/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 6.NS.B.4 (LCM & GCD)
- CCSS 4.OA.B.4 (Factor Pairs)
- TEKS 111.26.B.6.7.A (Prime Factor...)

Grades 4–9 Math & Pre-Algebra

GUIDING INQUIRY

Which prime blocks are shared by both towers, and which blocks are unique to one tower?

1 STEP 1: FACTOR & BUILD TOWERS 12 & 18

Decompose numbers 12 and 18 into prime building blocks:

- Tower 12: Stack [2] + [2] + [3] (2 × 2 × 3 = 12)
- Tower 18: Stack [2] + [3] + [3] (2 × 3 × 3 = 18)

2 STEP 2: FIND SHARED OVERLAP (THE GCD)

Stand Tower 12 and Tower 18 side-by-side. Identify the blocks they share in common (intersection):

- Tower 12 has: [2], [2], [3] I Tower 18 has: [2], [3], [3]

Shared Overlap: Exactly one [2] and one

3–4 STEPS 3 & 4: ASSEMBLE THE LCM TOWER

3. TAKE PRIME UNION: Take highest power of each prime: two [2]s, two [3]s.

4. STACK LCM TOWER: [2] + [2] + [3] + [3] = 36 (Summit: 36.20 mm)!

Calculator Verification: Press Factor Tree Icon ([PRIME] / [g][FACT]) and run 12 [ENTER] 18 [GCF] 12 [ENTER] 18 [f] [LCM] → 36.

5 CORE PRINCIPLE: CONSERVATION OF PRIMES: A × B = GCD × LCM

Multiply Tower 12 and Tower 18: you have three [2]s and three [3]s. Look at your GCD Tower ([2], [3]) and your LCM Tower ([2]², [3]²): together they contain the EXACT same prime blocks!

Therefore: A × B = GCD(A, B) × LCM(A, B) = 6 × 36 = 216!

★ EXTENSION INVESTIGATION: FIND LCM(8, 12)

Build Tower 8: [2]+[2]+[2] and Tower 12: [2]+[2]+[3]. Prime union takes three [2]s and one [3] → LCM = 24 (32.10mm)! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

PRIME FACTOR STUDIO — PACKING BENCH & QC SPEC

4-3-3-2 Stepped-Floor Layout, Pack-Up Verification & Parts Replacement Manifest

SKU: PRIME-01-PACK

REV a793ddd

WAREHOUSE FULFILLMENT & PACKING BENCH REFERENCE • REV a793ddd • 3x1x7U Stepped-Floor Storage Box with 42.0mm Wa… …ray

PACKING BENCH MANIFEST & REFILL REPLACEMENT PRICING

#	COMPONENT DESCRIPTION	QTY	COLOR / SPEC	CONTENT-ADDRESSED REF	REFILL
①	**Prime Block [2] (1U = 7.00mm vis)**	**24x**	Matte Blue	PRIM-BLCK-02	$2.49
②	**Prime Block [3] (1.58U = 11.10mm vis)**	**9x**	Muted Red	PRIM-BLCK-03	$2.49
③	**Prime Block [5] (2.32U = 16.25mm vis)**	**6x**	PLA White	PRIM-BLCK-05	$2.49
④	**Prime Block [7] (2.81U = 19.65mm vis)**	**4x**	PLA Tan	PRIM-BLCK-07	$2.49
⑤	**3x1x7U Stepped-Floor Storage Box**	**1x**	PLA Gray	GEN-PRIM-BOX7U	$14.50
⑥	**12-Socket Comparison Stage Lid**	**1x**	PLA White	GEN-PRIM-LID01	$7.50

TOTAL: 45 VERIFIED WORKING COMPONENTS

WAREHOUSE NO LOST PIECE GUARANTEE

Individual replacement blocks are permanently stocked in Mountain View, California.
Anti-arbitrage pricing: Order single replacement components anytime at stackcalc.io/parts.
Never retire an educational classroom kit due to a single misplaced component.

PACKING QC & WATERLINE PROTOCOL

FINAL BENCH INSPECTION

- ✓ All 43 pure prime blocks verified: 24x[2], 9x[3], …
- ✓ Harmonized 1U log datum: h(2)=7.00mm, K=23.2535 mm …
- ✓ Stepped floor sockets labeled 2 (x4), 3 (x3), 5 (x…
- ✓ All 12 columns level at 42.0mm (2^6=64) waterline;

PACKING WATERLINE NOTICE

SELF-CHECKING PACK-UP WATERLINE: Seat all 43 blocks into the 12 stepped-floor sockets (4 cols of 6x[2], 3 cols of 3x[3], 3 cols of 2x[5], 2 cols of 2x[7]) so all 12 columns level off at the 42.0mm waterline before snapping the lid shut.

CALIFORNIA LAB QC AUDIT

INSPECTION: PASSED

Inspector: Station 04 • Mountain View, CA
Lot Release: 2026-Q4-ARCHIVAL
Standards: ASTM F963 / EN71 / RoHS Lead-Free

FACTORY SPECIFICATION
Harmonized Modular Mathematics System • Manufactured with Pride in California, USA.

INSTITUTIONAL DIRECT
sales@stackcalc.io

PRIME FACTOR STUDIO — PACKING BENCH & QC SPEC

WAREHOUSE PACKING BENCH REFERENCE • 3D VIDEO QR • REFILL LOGISTICS

Classroom Pack-Up & Inventory Reference

SKU: PRIME-01-PACK

PACKING PROTOCOL

PACK-UP DEMO VIDEO

OPEN CAMERA & SCAN

stackcalc.io/products/prime-studio/

Watch 3D video demonstration

WAREHOUSE PACKING BENCH & BOX FLOOR WATERLINE AUDIT

1. SEAT 43 BLOCKS IN 4–3–3–2 SOCKETS

Pack 6x[2] in the 4 left sockets (Cols 0-1), 3x[3] in Front Cols 2-4, 2x[5] in Back Cols 2-4, and 2x[7] in Right Cols 5.

2. CHECK THE 42.0MM WATERLINE

Confirm all 12 packed columns level off flush at the interior 42.0mm (2^6=64) stepped-floor waterline.

3. TUCK CARD & SNAP LID SHUT

Place your reference card on top and snap the 12-socket stage lid onto the rim for rattle-free storage.

FULFILLMENT PORTAL

Interactive Packaging Bench Station • Warehouse Fulfillment Rev a793ddd.

REPLACEMENT MANIFEST

stackcalc.io/parts

SC-6 QUEST SCOREBOARD & +10 TOKEN KIT

Daily Multi-Method Mastery Scoreboard (Sockets 1–10, 100 & ? Badge)

SERIES 1 • CARD 10 / 12 • DOMAIN: MULTI-METHOD MASTERY & DAILY FLUENCY

CARD 10 / 12 • QUEST-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

ONE PROBLEM. MULTIPLE WAYS. SCORE YOUR MASTE

Confidence grows by solving in multiple ways.

Reward each approach with tactile score tokens:

1 Block = +10 (1st Way) • 2 Blocks = +20 (2nd Way)

Stacking a 2nd token rewards a 2nd solution method!

UNIVERSAL +10 QUARTER-BLOCK DATUM

- Token: Universal +10 Block = 10.25×10.25×7.0mm
- 1st Way: Seat 1 token in socket 1–10 (+10 pts)
- 2nd Way: Stack 2nd token on top (+20 pts total)

LID SOCKETS 1–10 & 100: 1 BLOCK = +10, 2 STA



+10 TOKENS

- 1 Block = +10 pts
- 2 Stacked = +20 pts

100 Mastery Tower:

Trade 10 blocks into Socket 1(

Matches SC-6 [¯][C] (?) Quiz

2. KIT PARTS INVENTORY (62 PIECES)

COMPONENT	QTY	SPEC	COLOR
1st-Way Tokens	30x	10.25 mm	Red+Gold Silk
2nd-Way Tokens	30x	10.25 mm	Glow Yellow
3x1 Storage Bin	1x	49.0 mm	PLA White
Scoreboard Lid	1x	Scoreboard	PLA White

TOTAL: 60 Universal +10 Tokens + 7U Bin + Lid = 62 Parts

DUAL-TONE +10 TOKEN PALETTE

- Interchangeable Tokens: All 60 quarter-blocks are +10 pts.
- Red+Gold Silk (30x): Seat for 1st Way solution (+10 pts).
- Glow-in-the-Dark Yellow (30x): Stack on top for 2nd Way (+20 pts)!
- High Visibility: Easy to tally scores at a glance.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Sweep all 60 universal +10 tokens directly into the open 3x1x7U bin, lay this reference card on top, and snap the Scoreboard Lid shut for silent, zero-rattle classroom storage.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!

Keep stored inside box lid!

FUNDAMENTAL THEOREM

Solving one problem in two independent ways builds deeper mastery than solving ten by rote.

STACKCALC ARCHIVAL INSTRUMENT LABS

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL

stackcalc.io/parts

DAILY PROTOCOL: 10 QUESTIONS, 2 WAYS & THE 100 MASTERY TOWER

How to Use Your SC-6 Quest Scoreboard with Daily Practice & Left-Shift [C] (?) Tutorial Mode

SERIES 1 • CARD 10 OF 12 • DOMAIN: MULTI-METHOD MASTERY & DAILY FLUENCY

CARD 10 / 12 • QUEST-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/quest-scoreboard/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS MP1 (Persevere & Verify)
- CCSS MP5 (Use Strategic Tools)
- SC-6 Built-In Scoring (+10 / +20 / ...)

Grades 3–8 Daily Math Fluency

GUIDING INQUIRY

Can you verify your textbook calculation using physical blocks or the SC-6 RPN stack to earn a second +10 token?

① STEP 1: SOLVE 1ST WAY (+10 TOKEN)

Place the Scoreboard Lid on your desk beside your worksheet.
When you solve Question 1 using your first method (Textbook, Blocks, or SC-6), seat one +10 token into Socket [1].

② STEP 2: TRY A 2ND WAY (+20 STACK)

Pick any question and check it a SECOND way:
• Built it with blocks? Check with SC-6 RPN keystrokes!
• Solved by hand? Prove it with Prime or Fraction blocks!

3–4 STEPS 3 & 4: BUILD THE [100] MASTERY TOWER

3. TRADE 10 TOKENS FOR 100: Every 10 tokens (100 points), stack them into a 10-block tower in Socket [100]!
4. LAUNCH SC-6 QUEST MODE: Press Left-Shift [f] then [C] (?) matching the Question Mark in a Circle socket.
Calculator Verification: Press Question Mark in a Circle Tutorial Icon ([f][?]) and run 8 [ENTER] 2 [+] 5 [+] [■] → 15

⑤ CORE PRINCIPLE: WHY TWO WAYS BEATS ROTE DRILL

Students who think they are 'bad at math' gain immediate confidence when physical blocks, textbook steps, and the SC-6 calculator all land on the exact same answer. Every 2-block stack
on your scoreboard is physical proof that mathematics is consistent and discoverable!

★ EXTENSION INVESTIGATION: EARN A 200-POINT DOUBLE MASTERY DAY

Solve all 10 daily problems using your 1st Way (100 pts), then verify 10 problems using a 2nd Way to stack 20 tokens (200 pts total)!

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

SC–6 QUEST SCOREBOARD — PACKING BENCH & QC SPEC

2-Piece Enclosure & Lid Inspection, 60-Token Count & Replacement Manifest

WAREHOUSE FULFILLMENT & PACKING BENCH REFERENCE • REV a793ddd • 3x1x7U Modular Storage Bin (No Inner Tray) + Sn

SKU: QUEST–01–PACK

REV a793ddd

PACKING BENCH MANIFEST & REFILL REPLACEMENT PRICING

#	COMPONENT DESCRIPTION	QTY	COLOR / SPEC	CONTENT-ADDRESSED REF	REFILL
①	**+10 Score Token — 1st Way (10.25mm)**	**30x**	Gold Silk	**QUEST–TOKN–GOLD**	**$1.88**
②	**+10 Score Token — 2nd Way (10.25mm)**	**30x**	Yellow	**QUEST–TOKN–GLOW**	**$1.88**
③	**+10 Score Token Refill Pack (10x)**	**1x**	Assorted	**QUEST–PACK–10**	**$6.75**
④	**3x1 Scoreboard Lid (1–10, 100, ?)**	**1x**	Technical White	**QUEST–LID–a793ddd**	**$9.00**
⑤	**3x1x7U Storage Enclosure Body**	**1x**	Technical White	**QUEST–BOX–a793ddd**	**$12.00**

TOTAL: 62 VERIFIED WORKING COMPONENTS

WAREHOUSE NO LOST PIECE GUARANTEE

Individual replacement blocks are permanently stocked in Mountain View, California.
Anti-arbitrage pricing: Order single replacement components anytime at stackcalc.io/parts.
Never retire an educational classroom kit due to a single misplaced component.

PACKING QC & WATERLINE PROTOCOL

FINAL BENCH INSPECTION

- ✓ 60 universal +10 tokens verified (30x Gold + 30x G
- ✓ Lid sockets 1–10 & 100 tested for smooth 0.40mm sl
- ✓ Question Mark in a Circle (?) badge crisp on lid &
- ✓ 2-piece snap-lock lid clicks firmly onto 3x1x7U op

PACKING WATERLINE NOTICE

SIMPLIFIED 2-PIECE DESIGN (NO INNER TRAY):
Pour all 60 universal +10 tokens into the open 3x1x7U storage bin, place this card on top, and snap the Scoreboard Lid shut.

CALIFORNIA LAB QC AUDIT

INSPECTION: PASSED

Inspector: Station 04 • Mountain View, CA
Lot Release: 2026-Q4-ARCHIVAL
Standards: ASTM F963 / EN71 / RoHS Lead-Free

FACTORY SPECIFICATION
Harmonized Modular Mathematics System • Manufactured with Pride in California, USA.

INSTITUTIONAL DIRECT
sales@stackcalc.io

SC-6 QUEST SCOREBOARD — PACKING BENCH & QC SPEC

WAREHOUSE PACKING BENCH REFERENCE • 3D VIDEO QR • REFILL LOGISTICS

Classroom Pack-Up & Inventory Reference

SKU: QUEST-01-PACK

PACKING PROTOCOL

PACK-UP DEMO VIDEO

OPEN CAMERA & SCAN

stackcalc.io/products/quest-scoreboard/

Watch 3D video demonstration

WAREHOUSE PACKING BENCH & BOX FLOOR WATERLINE AUDIT

1. VERIFY 60 UNIVERSAL +10 TOKENS

Confirm 30x Red+Gold Silk tokens and 30x Glow-in-the-Dark Yellow tokens rest inside the open 3x1x7U storage bin.

2. INSPECT SCOREBOARD LID SOCKETS

Verify sockets 1–10, the 100 Mastery Tower socket, and the Question Mark in a Circle (?) tutorial keycap badge.

3. TUCK CARD & SNAP LID SHUT

Place this reference card over the tokens and snap the 3x1 Scoreboard Lid onto the rim.

FULFILLMENT PORTAL

Interactive Packaging Bench Station • Warehouse Fulfillment Rev a793ddd.

REPLACEMENT MANIFEST

stackcalc.io/parts

NUMBER RACK (REKENREK) ARITHMETIC STUDIO

Dual-Row 20-Bead 5-and-10 Grouping & Subitizing Frame

SERIES 1 • CARD 11 / 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

MATHEMATICAL GUIDE

1. CONCEPT & GEOMETRIC PROOF

5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

Counting by ones is slow and error-prone. The Rekenrek organizes 20 beads into 5-anchors:
Row 1: 5 Red + 5 White • Row 2: 5 Red + 5 White
Sliding beads in groups turns math into subitizing!

CALIBRATED 102.0 MM SLIDER DATUM

- Beads: 10 Red + 10 White = 20 Beads (14.0 mm ⌀)
- Slider Rods: 2x 102.0 mm Octagonal Rods
- Upright Towers: 2x 49.0 mm Snap-Fit Towers (7U)

5-AND-10 REKENREK: 7 + 6 = 13



DOUBLES-PLUS-ONE

- Red: 5 + 5 = 10 (Instant T
- White: 2 + 1 = 3 Units

Sum = 10 + 3 = 13

Zero finger counting!

2. KIT PARTS INVENTORY (27 PIECES)

COMPONENT	QTY	SPEC	COLOR
Red Beads	10x	14.0 mm Bead	Crimson Red
White Beads	10x	14.0 mm Bead	Classic White
Slider Rods	2x	102.0 mm Rod	Matte Charcoal
Upright Towers	2x	49.0 mm	Matte Charcoal
Storage Tray	1x	Nesting Tray	Matte Charcoal
3x1 Storage Bin	1x	49.0 mm	Matte Charcoal
Stage Lid	1x	Slider Lid	Matte Charcoal

TOTAL: 20 Beads + 4 Frame Parts + Tray + Storage Bin + Lid = 27 Parts

DUAL-ROW 5-COLOR STRUCTURE

- 5 Red + 5 White per row: Instant 5-anchor visual subitizing.
- Both Rows Active: Models doubles, near-doubles, and bridging.
- Zero-Rattle Carry: Beads slide into nested tray slots.
- High Contrast: Crisp color separation across rows.

3. PACK-UP & SUPPORT

✓ 10-SECOND PACK-UP

Slide 10 red beads left and 10 white beads right on both rods. Place rods and towers into the nesting tray. Missing beads leave a visible gap. Snap lid shut.

✓ NO LOST PIECE GUARANTEE

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts
Order replacements anytime

✓ HANDS-ON TUTORIAL

Flip over for 5-step concrete lesson & live 3D Video QR!
Keep stored inside box lid!

FUNDAMENTAL THEOREM

Dual 5-and-10 grouping bridges perceptual subitizing directly into formal arithmetic operations.

STACKCALC ARCHIVAL INSTRUMENT LABS
Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

OFFICIAL REFILL PORTAL
stackcalc.io/parts

LESSON GUIDE: DUAL-ROW BRIDGING & DOUBLES PLUS ONE

Step-by-Step Hands-On Tutorial: 7 + 6 = 13 via 5-and-5 Doubles

SERIES 1 • CARD 11 OF 12 • DOMAIN: 5-AND-10 GROUPING & DUAL-ROW ARITHMETIC

CARD 11 / 12 • RACK-01

LESSON TUTORIAL

SCAN FOR 3D VIDEO DEMO

OPEN CAMERA & SCAN

stackcalc.io/products/number-rack/

Zero app install • iPad & Chromebook

CURRICULAR ALIGNMENT

- CCSS 1.OA.C.6 (Addition Fluency)
- CCSS 2.OA.B.2 (Mental Math)
- TEKS 111.3.b.3.C (Compose/Deco...)

Grades K–3 Subitizing & Addition

GUIDING INQUIRY

When you slide 7 beads on top and 6 on bottom, how many red beads are on the left side altogether?

1 STEP 1: SLIDE 7 BEADS ON TOP ROW

Push all beads to the right side of the rack (resting zero).
Slide 7 beads to the left on the top rod in one push:
All 5 red beads + 2 white beads = 7 beads!
Notice: 5-anchor lets you see 7 instantly

2 STEP 2: SLIDE 6 BEADS ON BOTTOM ROW

Slide 6 beads to the left on the bottom rod in one push:
All 5 red beads + 1 white bead = 6 beads!
We want to find 7 + 6 without counting one-by-one.

3–4 STEPS 3 & 4: GROUP DOUBLES & REGROUP

3. GROUP RED BEADS: Look at the red beads on the left: 5 + 5 = 10 (Instant Ten)!
4. ADD WHITE BEADS: Look at the white beads: 2 + 1 = 3 units.
Total: 10 + 3 = 13! Therefore: 7 + 6 = 13!
Calculator Verification: Press Number Rack Subitizing Icon ([▦]) and run [7] [ENTER] [6] [+] → 13.0000.

5 CORE PRINCIPLE: DOUBLES MAKE MENTAL MATH EFFORTLESS

Instead of counting 6 fingers past 7, the Rekenrek decomposes 7 + 6 into (5 + 2) + (5 + 1). The two groups of 5 red beads instantly make 10, leaving 3 white beads. Addition becomes instantaneous visual pattern recognition rather than tedious memorization!

★ EXTENSION INVESTIGATION: SOLVE: 8 + 7 USING 5-AND-5

Slide 8 beads on top (5 red + 3 white) and 7 on bottom (5 red + 2 white). Group the 10 red beads first! What is the sum? (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.